

Solid LB Preparation

Lysogeny broth, Luria-Bertani, medium is a common medium for growing bacteria, such as *E. coli* in liquid or on solid agar plates. This protocol is for 500 mL of LB media. To prepare liquid LB broth, omit addition of agar.

Reagents

✓	Name	Quantity	Location/Notes
☐	ddH ₂ O	500mL	
☐	LB powder	12.5g	5g tryptone 5g sodium chloride 2.5g yeast extract
☐	Agar powder	7.5g	Can use bacto-agar but is expensive
☐	Ampicillin	0.05g	If using [2]

Equipment

✓	Name	Quantity	Location/Notes
☐	Analytical scale		
☐	Autoclave		Training required
☐	1L autoclavable bottle		
☐	Sterile Agar plates	20	
☐	Parafilm		
☐	Sterile bags		

☐ Stir bar		
☐ BSC	Training required	

Instructions

1. Add the following to the 1L bottle, swirling to mix:

Reagent Tracker	Quantity
☐ LB powder	12.5g
☐ ddH2O	to 450mL

2. If making **solid media**, add the following, note that the Bacto-agar will not dissolve.

Reagent Tracker	Quantity
☐ Bacto-agar	15g

3. Add ddH2O until a final volume of 500mL is reached.

Reagent Tracker	Quantity
☐ ddH2O	to 500mL

4. Add aluminum to the top of the flask to cover but allow for air flow. Autoclave on a liquid cycle (121°C for 15min)
5. Allow flask to cool until about 50°C
6. Add **antibiotic**, if using:

Reagent Tracker	Quantity
☐ Ampicillin	0.05g

7. If making solid media, in a BSC, unstack sterile plates
8. Pour the media (about 25mL) into the plates and allow to cool until solid (30min - 1h)
9. Seal plates with parafilm, label, and place in sterile bags to store at 4°C

Notes here

1. Plasmid contains ampicillin resistance, though not sure if using
2. Store upside-down to reduce condensation
3. DO NOT RE-AUTOCLAVE, can try to use a water bath to re-melt solidified agar

References

[1] 2025 cycle LB Media Preparation

[2]

<https://www.protocols.io/view/antibiotic-concentrations-for-lb-agar-plates-bghkjt4w.pdf>

<https://www.protocols.io/view/LB-Agar-Plates-j8epv5xng1bz/v1>